

Run 08 | OME

Global (including India)

IV + controls

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 100,087 (7.45%) **Case time** 49m 06s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9784). Highest F1: Random Forest (0.7140). For Random Forest, adding controls changed ROC-AUC by +0.0027 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9784	0.7444	95.04	62.63	83.03	0.7140	2m 35s
XGBoost	0.9783	0.7361	95.03	62.59	82.81	0.7129	22s
CatBoost	0.9779	0.7319	94.97	62.26	82.54	0.7098	2m 41s
H2O GBM	0.9766	0.7138	94.55	59.23	86.06	0.7017	1m 06s
Elastic Net	0.9745	0.6844	94.31	58.14	84.69	0.6894	15m 12s
Lasso	0.9745	0.6843	94.31	58.14	84.69	0.6895	11m 24s
Logistic Regression	0.9744	0.6834	94.32	58.18	84.80	0.6901	1m 56s
Ridge	0.9738	0.6748	94.35	58.32	84.87	0.6913	7m 53s
Decision Tree	0.9707	0.6255	93.99	56.51	84.24	0.6764	15s
Naive Bayes	0.8794	0.2899	86.64	31.42	67.00	0.4278	4m 51s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	30,148.96	35.49
2	Established business owner	11,542.02	13.59
3	Occupational status	4,779.09	5.63
4	Age	4,001.56	4.71
5	Household size	1,798.27	2.12
6	Self-efficacy	1,763.61	2.08
7	Knowing entrepreneurs	1,665.24	1.96
8	Opportunity perception	1,427.90	1.68

Share is the percentage of total forest impurity importance across all 66 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Safety and security	1,259.31	1.48
2	Living conditions	1,189.26	1.40
3	Education	1,169.09	1.38

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 08_obj2_global_iv_cv_OME. Scores and importance are taken directly from saved CSV outputs.